

CLASS XI BIOLOGY

HUMAN PHYSIOLOGY

NCERT PAGES 240–251

CHAPTER 19: CHEMICAL COORDINATION AND INTEGRATION

Complete exam-oriented comprehensive study guide covering endocrine glands, hormonal regulation, target tissues, feedback loops, and molecular mechanisms of action.

DOCUMENT TYPE

Master Notes

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CHAPTER MIND MAP — ENDOCRINE SYSTEM AT A GLANCE

HIERARCHICAL STRUCTURE

1. Master Control



- **Hypothalamus:** Basal diencephalon, Releasing & Inhibiting factors.
- **Pituitary:** Adenohypophysis (Pars distalis & intermedia) + Neurohypophysis.

2. Metabolic Glands



- **Thyroid:** T₃, T₄, Thyrocalcitonin (TCT), BMR regulation.
- **Parathyroid:** PTH (Hypercalcemic balance).
- **Pineal:** Melatonin (Diurnal rhythms).

3. Abdominal / Gonadal



- **Adrenal:** Cortex (Corticoids) + Medulla (Catecholamines - 3F).
- **Pancreas:** Insulin (β -cells) & Glucagon (α -cells).
- **Gonads:** Testis (Androgens) & Ovary (Estrogen/Progesterone).

4. Non-Endocrine & Action



- **Heart/Kidney/GIT:** ANF, Erythropoietin, Gastrin, CCK, Secretin, GIP.
- **Mechanism:** Membrane receptors (2nd messengers) vs Intracellular (Genome).

19.1 ENDOCRINE GLANDS AND HORMONES

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Neural Coordination vs. Endocrine Coordination

The neural system provides a **point-to-point rapid coordination** among organs. However, neural coordination is *fast but short-lived*. Nerve fibers do **not innervate all cells** of the body, and cellular functions require continuous regulation.

- ✓ **Core Principle:** The Neural System and Endocrine System jointly coordinate and regulate all physiological functions in the body.

Endocrine Glands & Definition of Hormones

Endocrine glands **lack ducts** and are hence called **ductless glands**. Their secretions are poured directly into the bloodstream and carried to target organs.

CLASSICAL DEFINITION

Chemical produced by endocrine glands, released into blood, transported to distantly located target organs.

CURRENT SCIENTIFIC DEFINITION

Hormones are **non-nutrient chemicals** which act as **intercellular messengers** and are produced in **trace amounts**.



Comparative Endocrinology Fact:

Invertebrates possess very simple endocrine systems with very few hormones. In contrast, **Vertebrates** possess a large number of chemicals acting as hormones to maintain complex homeostatic coordination.

Organised Endocrine Bodies in Humans: Pituitary Pineal Thyroid Parathyroid Thymus Adrenal Pancreas Gonads (Testis & Ovary)

Note: Diffused hormone-producing tissues are also located in Gastrointestinal tract, Liver, Kidney, and Heart.

 19.2.1 The Hypothalamus

Basal Forebrain

Anatomical Location & Structure

- Forms the **basal part of diencephalon** (forebrain).
- Contains groups of neurosecretory cells called **nuclei**.
- Nuclei synthesize and secrete hypothalamic hormones regulating the pituitary gland.

Types of Hypothalamic Hormones

1. Releasing Hormones: Stimulate pituitary secretions.

Example: GnRH (Gonadotrophin Releasing Hormone) → stimulates pituitary synthesis & release of gonadotrophins.

2. Inhibiting Hormones: Inhibit pituitary secretions.

Example: Somatostatin → inhibits release of Growth Hormone (GH) from pituitary.

Dual Neural & Vascular Regulation Route

Anterior Pituitary: Hypothalamic neurons → axonal endings → **Portal Circulatory System** → regulates anterior pituitary.

Posterior Pituitary: Under **direct neural regulation** of hypothalamus.

  **Easy Way to Remember Route:** "**Anterior Portal, Posterior Neural**" — Anterior Pituitary is controlled via blood (Portal Circulation), while Posterior Pituitary is directly wired via Axons (Neural Control).

Anatomical Location & Anatomical Divisions

Located in a bony cavity called **sella tursica** of the sphenoid bone and attached to the hypothalamus by a **stalk** (Infundibulum).

1. Adenohypophysis

- **Pars Distalis (Anterior Pituitary):** Secretes GH, PRL, TSH, ACTH, LH, and FSH.
- **Pars Intermedia:** Secretes MSH (Melanocyte Stimulating Hormone). *Note: In humans, pars intermedia is almost merged with pars distalis.*

2. Neurohypophysis (Pars Nervosa / Posterior Pituitary)

- **Stores and releases 2 hormones:** Oxytocin and Vasopressin (ADH).
- *Crucial Fact:* These hormones are **synthesized by the Hypothalamus** and transported axonally to neurohypophysis!

Hormone Name	Source Region	Primary Biological Function / Mechanism	Associated Clinical Disorder / Condition
Growth Hormone (GH)	Pars Distalis	Stimulates overall body growth, protein synthesis, tissue repair, and cell elongation.	• Hypersecretion (Children): Gigantism. • Hyposecretion (Children): Pituitary Dwarfism. • Hypersecretion (Adults/Middle Age): Acromegaly (Severe facial disfigurement, premature death if unchecked).
Prolactin (PRL)	Pars Distalis	Regulates growth of mammary glands and formation of milk in them.	Disruption leads to inadequate milk production post-parturition.
Thyroid Stimulating Hormone (TSH)	Pars Distalis	Stimulates synthesis and secretion of thyroid hormones (T ₃ , T ₄) from thyroid gland.	Secondary Hypothyroidism or Hyperthyroidism.
Adrenocorticotrophic Hormone (ACTH)	Pars Distalis	Stimulates synthesis and secretion of steroid hormones called glucocorticoids from adrenal cortex.	Altered carbohydrate metabolism and steroid imbalances.

Hormone Name	Source Region	Primary Biological Function / Mechanism	Associated Clinical Disorder / Condition
Luteinizing Hormone (LH)	Pars Distalis (Gonadotrophin)	• Males: Stimulates synthesis and secretion of androgens from Leydig cells. • Females: Induces ovulation of Graafian follicles and maintains corpus luteum.	Gonadal dysfunction & infertility.
Follicle Stimulating Hormone (FSH)	Pars Distalis (Gonadotrophin)	• Males: Regulates spermatogenesis along with androgens. • Females: Stimulates growth and development of ovarian follicles.	Impaired gametogenesis.
Melanocyte Stimulating Hormone (MSH)	Pars Intermedia	Acts on melanocytes (melanin-containing cells) and regulates skin pigmentation.	Pigmentation abnormalities.
Oxytocin	Neurohypophysis (Hypothalamic origin)	Acts on smooth muscles; causes vigorous uterine contractions during childbirth and milk ejection from mammary glands.	Sluggish labor, delayed milk ejection.
Vasopressin / Anti-diuretic Hormone (ADH)	Neurohypophysis (Hypothalamic origin)	Acts mainly at kidneys to stimulate resorption of water & electrolytes by distal tubules, reducing diuresis.	Diabetes Insipidus: ADH deficiency leads to inability to conserve water, causing massive urine output, severe dehydration, and thirst.

⚠ Special Focus: Acromegaly

Excess GH secretion in adults (middle age) results in severe disfigurement (especially face). Hard to diagnose early, often undetected for years until external features change noticeably. Can cause serious complications and premature death.

🔑 Special Focus: Diabetes Insipidus

Impairment affecting synthesis or release of ADH results in diminished ability of kidneys to conserve water → excessive water loss (Diuresis) and dehydration. *Do not confuse with Diabetes Mellitus!*

Location: Dorsal side of forebrain.

Hormone Secreted: **Melatonin**

Functions of Melatonin:

- Regulates **24-hour (diurnal) rhythm** of the body.
- Maintains **sleep-wake cycle** and **body temperature**.
- Influences **metabolism** and **pigmentation**.
- Influences **menstrual cycle** and **defense capability**.

💡 Memory Trick: **Melatonin = Midnight Rhythm** (Sleep-Wake Balance)

Anatomy & Histology

- Composed of **two lobes** located on either side of the trachea.
- Interconnected with a thin flap of connective tissue called **Isthmus**.
- Composed of **follicles** and **stromal tissues**.
- Follicular cells enclose a cavity and synthesize T_4 & T_3 .

Hormones Synthesized

1. **Tetraiodothyronine (T_4)** / Thyroxine.
 2. **Triiodothyronine (T_3)**.
 3. **Thyrocalcitonin (TCT)**: Protein hormone regulating blood calcium levels (hypocalcemic).
- *Iodine is essential for normal rate of T_3 / T_4 synthesis!*

Thyroid Disorders Comparison:

Condition	Cause	Clinical Features & Effects
Hypothyroidism (Simple Goitre)	Deficiency of iodine in diet	Enlargement of thyroid gland.
Hypothyroidism during Pregnancy	Low thyroid levels in maternal body	Defective development of growing baby leading to Cretinism (stunted growth, mental retardation, low IQ, abnormal skin, deaf-mutism).
Hypothyroidism in Adult Women	Low hormonal secretion	Irregular menstrual cycles.
Hyperthyroidism / Graves' Disease	Thyroid cancer or development of thyroid nodules	Exophthalmic Goitre: Thyroid enlargement, protrusion of eyeballs, increased Basal Metabolic Rate (BMR), weight loss.

Major Functions of Thyroid Hormones (T_3 , T_4): (1) Regulation of Basal Metabolic Rate (BMR). (2) Support RBC formation (Erythropoiesis). (3) Control carbohydrate, protein & fat metabolism. (4) Maintain water and electrolyte balance.

19.2.5 PARATHYROID GLAND & CALCIUM HOMEOSTASIS

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Anatomy & Secretion

- **4 parathyroid glands** present on back side of thyroid gland (1 pair per lobe).
- Secretes peptide hormone: **Parathyroid Hormone (PTH)**.
- Secretion regulated by circulating blood Ca^{2+} levels.

PTH Mechanism (Hypercalcemic)

- **Increases blood Ca^{2+} levels** (Hypercalcemic hormone).
- Stimulates **bone resorption** (dissolution/demineralisation).
- Stimulates Ca^{2+} reabsorption by renal tubules.
- Increases Ca^{2+} absorption from digested food.

ANTAGONISTIC BLOOD Ca^{2+} REGULATION EQUATION

PTH (Parathyroid)
Raises Blood Ca^{2+}



TCT (Thyrocaltitonin)
Lowers Blood Ca^{2+}

Location: Lobular structure located between lungs behind sternum on ventral side of aorta.

Hormone Secreted: **Thymosins** (Peptide hormones).

Dual Immune Roles:

1. **Cell-Mediated Immunity:** Promotes differentiation of **T-lymphocytes**.
2. **Humoral Immunity:** Promotes production of **antibodies**.

⚠️ Age-Related Degeneration Fact:

Thymus degenerates in old individuals → decreased production of thymosins → immune responses of old persons become weak.

19.2.7 THE ADRENAL GLAND — CORTEX & MEDULLA

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One pair of adrenal glands, located anteriorly above each kidney. Composed of two distinct tissue zones: Central **Adrenal Medulla** and Outer **Adrenal Cortex**.

⚡ Adrenal Medulla (Inner Zone)

Emergency / 3F Gland

Hormones Secreted: *Adrenaline (Epinephrine)* and *Noradrenaline (Norepinephrine)*. Commonly called **Catecholamines**.

Emergency Responses ("Fight or Flight"):

- Increases alertness, pupillary dilation, piloerection (raising of hair), and sweating.
- Increases heart rate, strength of heart contraction, and rate of respiration.
- Stimulates breakdown of glycogen ($\rightarrow$ increased blood glucose).
- Stimulates breakdown of lipids and proteins.

🏠 Adrenal Cortex (Outer Zone)

3 Steroid Layers

Layers (Outside to Inside):

Zona Glomerulosa (Outer) → Zona Fasciculata (Middle) → Zona Reticularis (Inner)

1. Glucocorticoids (Main: Cortisol):

- Stimulates gluconeogenesis, lipolysis, proteolysis.
- Inhibits cellular uptake and utilization of amino acids.
- Maintains cardiovascular & kidney functions.
- Produces **anti-inflammatory reactions** & suppresses immune response.
- Stimulates RBC production.

2. Mineralocorticoids (Main: Aldosterone):

- Acts at renal tubules: Reabsorption of Na^+ & water; excretion of K^+ & phosphate ions.
- Maintains electrolytes, body fluid volume, osmotic & blood pressure.

3. Androgenic Steroids: Secretes small amounts; plays role in growth of axial, pubic & facial hair during puberty.



Clinical Disease Focus: Addison's Disease

Underproduction of hormones by the adrenal cortex alters carbohydrate metabolism causing acute weakness and fatigue leading to **Addison's Disease**.

19.2.8 PANCREAS — COMPOSITE GLAND & GLUCOSE HOMEOSTASIS

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Pancreas is a **composite (heterocrine/amphicrine) gland** acting as both exocrine and endocrine gland. Endocrine portion consists of **Islets of Langerhans** (1 to 2 million islets, representing only 1–2% of total pancreatic tissue).

Cell Type	Hormone Secreted	Primary Target Cells & Metabolic Actions	Overall Effect on Blood Glucose
α-cells (Alpha cells)	Glucagon (Peptide hormone)	• Acts on Hepatocytes (liver cells). • Stimulates Glycogenolysis (breakdown of glycogen to glucose). • Stimulates Gluconeogenesis. • Reduces cellular glucose uptake and utilization.	HYPERGLYCEMIC HORMONE (Increases blood glucose level)

Cell Type	Hormone Secreted	Primary Target Cells & Metabolic Actions	Overall Effect on Blood Glucose
β-cells (Beta cells)	Insulin (Peptide hormone)	- Acts on Hepatocytes and Adipocytes. - Enhances cellular glucose uptake and utilization. - Rapid movement of glucose from blood to target cells. - Stimulates Glycogenesis (conversion of glucose to glycogen).	HYPOGLYCEMIC HORMONE (Decreases blood glucose level)

Dual Feedback Balance Equation

High Blood Glucose → **[Insulin (β-cells)]** → Glycogenesis → Normal Blood Glucose

Low Blood Glucose → **[Glucagon (α-cells)]** → Glycogenolysis → Normal Blood Glucose

Clinical Disorder: Diabetes Mellitus

Prolonged hyperglycemia leads to **Diabetes Mellitus**, associated with loss of glucose through urine (glucosuria) and formation of harmful compounds called **ketone bodies** (ketonuria). Treated with *insulin therapy*.

19.2.9 TESTIS (MALE GONAD)

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Located in the scrotal sac (outside abdomen). Functions as primary sex organ and endocrine gland. Composed of seminiferous tubules and interstitial tissue.

Leydig / Interstitial Cells:

Present in intertubular spaces; produce a group of hormones called **Androgens** (mainly **Testosterone**).

Functions of Androgens:

- Regulate development, maturation, and functions of male accessory sex organs (epididymis, vas deferens, seminal vesicles, prostate, urethra).
- Stimulate muscular growth, facial & axillary hair, aggressiveness, low pitch voice.
- Stimulate **Spermatogenesis** (sperm formation).
- Act on CNS to influence male sexual behavior (**libido**).

- Produce **anabolic (synthetic) effects** on protein and carbohydrate metabolism.

19.2.10 OVARY (FEMALE GONAD)

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Pair of ovaries located in the abdomen. Primary female sex organ producing one ovum per menstrual cycle and two groups of steroid hormones: **Estrogen** and **Progesterone**.

1. Estrogen

Source: Growing ovarian follicles.

- Growth of female secondary sex organs.
- Secondary sex characters (high pitch voice).
- Mammary gland development & female sexual behavior.

2. Progesterone

Source: **Corpus Luteum** (ruptured follicle remnant).

- **Supports Pregnancy.**
- Acts on mammary glands to form **alveoli** (milk-storing sacs) and milk secretion.

19.3 HORMONES OF HEART, KIDNEY AND GASTROINTESTINAL TRACT

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Organ / Tissue	Hormone Secreted	Chemical Nature	Specific Biological Function & Target Action
Heart (Atrial Wall)	Atrial Natriuretic Factor (ANF)	Peptide	Causes dilation of blood vessels to decrease blood pressure when BP increases.
Kidney (Juxtaglomerular / JG Cells)	Erythropoietin	Peptide	Stimulates Erythropoiesis (formation of Red Blood Cells / RBCs).
Gastrointestinal Tract (GIT)	Gastrin	Peptide	Acts on gastric glands → stimulates secretion of Hydrochloric Acid (HCl) and Pepsinogen .
	Secretin	Peptide	Acts on exocrine pancreas → stimulates secretion of water and bicarbonate ions .
	Cholecystikinin (CCK)	Peptide	Acts on Pancreas (→ pancreatic enzymes) and Gall Bladder (→ bile juice).

Organ / Tissue	Hormone Secreted	Chemical Nature	Specific Biological Function & Target Action
	Gastric Inhibitory Peptide (GIP)	Peptide	Inhibits gastric secretion and motility.
Other Non-Endocrine Tissues	Growth Factors	Peptide/Protein	Essential for normal growth of tissues and their repairing/regeneration.

🧠💡 **Easy Way to Remember GIT Secretions:** "Gastrin = Acid | Secretin = Bicarbonate | CCK = Enzymes & Bile | GIP = Stop Acid"

19.4 MECHANISM OF HORMONE ACTION

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4 Chemical Classes of Hormones (NCERT Classification):

1. Peptide / Protein

Insulin, Glucagon, Pituitary hormones, Hypothalamic hormones.

2. Steroids

Cortisol, Testosterone, Estradiol, Progesterone.

3. Iodothyronines

Thyroid hormones (T_3 , T_4).

4. Amino-acid Derivatives

Epinephrine (Adrenaline).

A. Water-Soluble / Protein Hormones

Membrane-Bound Receptor

Hormones (e.g., **FSH**, **LH**, **Insulin**) interact with **membrane-bound receptors** on cell surface. They **do NOT enter target cell**.

- Hormone + Receptor → **Hormone-Receptor Complex**
- Generates **Response 1**
- Generates **Second Messengers** (Cyclic AMP, IP_3 , Ca^{2+})
- Biochemical responses → Physiological responses (e.g., ovarian growth).

B. Lipid-Soluble / Steroid Hormones

Intracellular / Nuclear Receptor

Hormones (e.g., **Estrogen, Cortisol, Iodothyronines**) cross membrane & bind to **intracellular receptors** (mostly nuclear).

1. Hormone passes membrane → Binds Intracellular Receptor
2. **Hormone-Receptor Complex** interacts with **Genome**
3. Regulates **Gene Expression / Chromosome Function**
4. Transcription/Translation → Proteins → Physiological effects (tissue growth/differentiation).



FINAL REVISION — HIGH-YIELD EXAM SUMMARY

Quick Revision

★ Emergency / 3F Hormones

Adrenaline & Noradrenaline (Catecholamines) from Adrenal Medulla respond to stress/emergencies by increasing heart rate, respiration, glycogenolysis, pupil dilation.

★ Antagonistic Pairs

1. **Insulin** (Hypoglycemic) vs **Glucagon** (Hyperglycemic).
2. **PTH** (Hypercalcemic) vs **TCT** (Hypocalcemic).

★ Key Clinical Disorders

- **Acromegaly:** Excess GH in adults.
- **Cretinism:** Hypothyroidism during pregnancy.
- **Diabetes Insipidus:** ADH deficiency.
- **Addison's Disease:** Adrenal Cortex hyposecretion.

★ Diurnal Rhythm Regulator

Melatonin from Pineal gland controls 24-hr diurnal rhythm, sleep-wake cycles, body temperature, metabolism, and immunity.

★ Immune System Gland

Thymus secretes Thymosins for T-lymphocyte differentiation (Cell-mediated) and antibody production (Humoral). Degenerates in old age.

★ Second Messengers

Generated by protein/peptide hormones binding to cell surface receptors: **Cyclic AMP (cAMP), IP₃, and Ca²⁺**.



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